Project lmeca2660 : chemical reaction under the action of an impinging jet

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In this project, it is proposed to study numerically the flow involved by the "burning of a solid fuel" (e.g. coal) under the action of an incident hot fluid jet, and also with natural convection effects. The domain has a rectangular shape of sides $L_x \times L_y$ (see Fig. 1), and the height is taken as $L_y = 2L_x$. The fluid enters the domain through an opening of width $L_{in} = \frac{Ly}{6}$; it lower part starts at the height $y_1 = \frac{L_{in}}{4}$. The outlet has a length $L_{out} = L_{in}$, centered in the middle of the top surface. Three species are involved: A, B and an inert I, with respective concentrations C_A, C_B and C_I (in $[\frac{mol}{m^3}]$). The reaction taking place at the bottom wall is A + coal $\longrightarrow$ B.

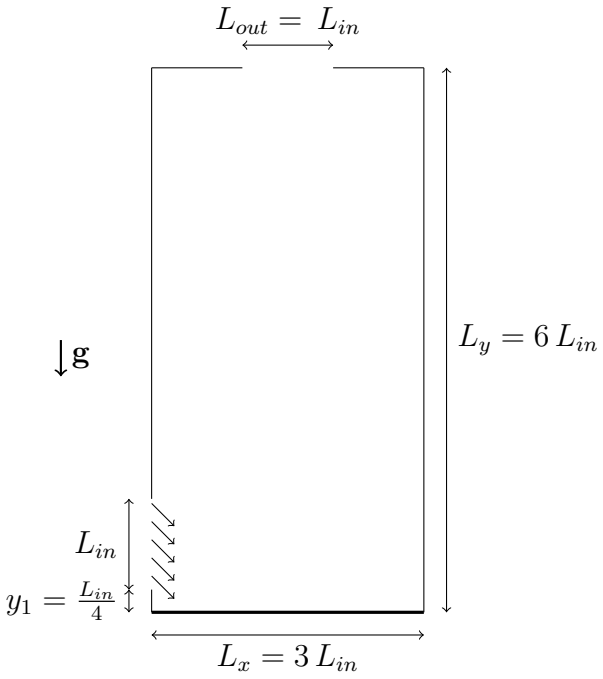


Figure 1: Configuration under study

Inlet The fluid is air that enters the domain as a jet at a uniform temperature T_m. The fluid in the domain is initially at a temperature $T_0 = 0.6\,T_m$. The velocity profile at the

inlet is taken as a Poiseuille flow, with an incident angle of $\theta = 45°$. The flow rate is Q (in $[\frac{m^3}{m.s}]$) and the reference velocity, U, is here defined as $U = Q/L_{in}$. The composition in the jet corresponds to that of the initial state in the domain: $C_A = 0.21\, C_{tot}$ (= typical oxygen concentration in air) and $C_I = 0.79\, C_{tot}$ (= typical nitrogen concentration in air), with C_{tot} the total concentration. The product concentration (carbon dioxide) is close to zero (not exactly as there is some carbone dioxide in air): we here simplify and take $C_B = 0$.

Walls A no-slip condition ($u_\Gamma = v_\Gamma = 0$) is to be imposed along the walls. On the side and top walls, no-flux conditions are assumed for the temperature and species, i.e. $\left.\frac{\partial T}{\partial n}\right|_\Gamma = \left.\frac{\partial C_i}{\partial n}\right|_\Gamma = 0$, for species i = 1,2,3. At the bottom wall, the consumption of A to produce B is modeled as a Robin boundary condition: $q_B|_\Gamma = -\, q_A|_\Gamma = K\, C_A$, with $q_i = -D_i \left.\frac{\partial C_i}{\partial y}\right|_\Gamma$ the molar flux of species i (in $[\frac{mol}{m^2.s}]$), D_i the mass diffusivity coefficient of species i (in $[\frac{m^2}{s}]$), and K the reaction rate constant. The latter is modeled using an Arrhenius law: $K = K(T) = K_0 \exp(-\frac{E}{RT})$ (in $[\frac{m}{s}]$), R being the perfect gas constant while K_0 and E are two reaction-related parameters. The heat generated by the reaction is assumed to be released entirely towards the flow, i.e. we neglect the conductive flux through the bottom wall. The heat flux density released at the bottom (in $[\frac{W}{m^2}]$) is then obtained as $q = -k \left.\frac{\partial T}{\partial y}\right|_\Gamma = q_B\, (\Delta H)$ with ΔH the molar enthalpy of the reaction (in $[\frac{J}{mol}]$).

Outflow A "natural" outflow condition is used for the fields at the outlet:

$$\frac{\partial u}{\partial y} = 0, \tag{1}$$

$$\frac{\partial v}{\partial t} + U\frac{\partial v}{\partial y} = 0, \tag{2}$$

$$\frac{\partial T}{\partial t} + U\frac{\partial T}{\partial y} = 0, \tag{3}$$

$$\frac{\partial C_i}{\partial t} + U\frac{\partial C_i}{\partial y} = 0, \tag{4}$$

Note: You might need to correct the outflow velocity in order to satisfy the flow rate conservation $\left(\int_{L_{in}} u\, dy = \int_{L_{out}} v\, dx\right)$ at each time step before solving the Poisson equation.

Equations and parameters of the problem We will make use of the Boussinesq approximation: the change of density is only taken into account via a buoyancy term in the momentum equation, while the flow is still assumed incompressible. Moreover, the

viscous dissipation will be neglected in the energy equation.

The equations to solve are then obtained as:

$$\nabla \cdot \mathbf{v} = 0 \tag{5}$$

$$\frac{D\mathbf{v}}{Dt} = -\nabla P + \nu \nabla^2 \mathbf{v} - \beta(T - T_m)\mathbf{g} \tag{6}$$

$$\frac{DT}{Dt} = \alpha \nabla^2 T \tag{7}$$

$$\frac{DC_i}{Dt} = D_i \nabla^2 C_i \tag{8}$$

with $\nu = \dfrac{\mu}{\rho_m}$ the kinematic viscosity, $P = \dfrac{(p - p_{ref}) + \rho_m g y}{\rho_m}$ the reduced kinematic pressure (p_{ref} is any reference pressure and $\rho_m = \rho(T_m)$), $\mathbf{g} = -g\hat{\mathbf{e}}_\mathbf{y}$ the gravitational acceleration, $\beta = -\dfrac{1}{\rho_m}\dfrac{\partial \rho}{\partial T}\bigg|_{T_m}$ the fluid expansion coefficient, and $\alpha = \dfrac{k}{\rho_m \, c_p}$ the thermal diffusivity.

The Reynolds number based on the inlet flow $Re = \dfrac{Q}{\nu} = 250$. The fluid Prandtl number is taken as $Pr = \dfrac{\nu}{\alpha} = 0.70$. The Schmidt number is supposed identical for all species: $D_i = D$ and $Sc = \dfrac{\nu}{D} = 0.10$. The Grashof number is $Gr = \dfrac{\beta \, g \, T_m \, (L_{in})^3}{\nu^2} = 2.0 \cdot 10^5$. The reaction-related parameters are such that $\dfrac{K_0 \, L_{in}}{D} = 1.0 \cdot 10^{10}$, $\dfrac{E}{R \, T_m} = 20$ and $\dfrac{\Delta H C_{tot} D}{k \, T_m} = 2.0$.

Flow solving

In order to integrate Eqns. (5) to (8), we use a two-step projection scheme combined with a MAC mesh. The convective terms are integrated using the Adams-Bashfort scheme of order 2 (with the explicit Euler scheme for the first step) while the diffusive terms are integrated using the explicit Euler scheme. Hence, the numerical scheme is:

$$\begin{aligned} \frac{(\mathbf{v}^* - \mathbf{v}^n)}{\Delta t} &= -\frac{1}{2}\left(3\,\mathbf{H}_h^n - \mathbf{H}_h^{n-1}\right) - \nabla_h P^n + \nu \nabla_h^2 \mathbf{v}^n \\ &\quad -\beta(T^n - T_m)\mathbf{g} \end{aligned} \tag{9}$$

$$\nabla_h^2 \Phi = \frac{1}{\Delta t} \nabla_h \cdot \mathbf{v}^* \tag{10}$$

$$\frac{(\mathbf{v}^{n+1} - \mathbf{v}^*)}{\Delta t} = -\nabla_h \Phi \tag{11}$$

$$P^{n+1} = P^n + \Phi \tag{12}$$

$$\frac{(T^{n+1} - T^n)}{\Delta t} = -\frac{1}{2}\left(3\,H_{T,h}^n - H_{T,h}^{n-1}\right) + \alpha \nabla_h^2 T^n \tag{13}$$

$$\frac{(C_i^{n+1} - C_i^n)}{\Delta t} = -\frac{1}{2}\left(3\,H_{C_i,h}^n - H_{C_i,h}^{n-1}\right) + D \nabla_h^2 C_i^n \tag{14}$$

where $\mathbf{H}_h$ is the disctretised convective term of the momentum equation, $H_{T,h}$ the discretised convective term of the energy equation and $H_{C_i,h}$ the discretised convective term of the species i transport equation.

The concentration of the inert species is not modified during the simulation. Indeed, one mole of A consumed releases one mole of B and the initial and inlet compositions are identical. Using mass conservation, we can write: $C_A + C_B = C_{tot} - C_I = 0.21 C_{tot}$. Hence, only one species transport equation has to be solved (e.g. A) and the concentration of the second species (e.g. B) is obtained straightforwardly.

We will here use a mesh with $\Delta x = \Delta y = h$, where h fulfils a constraint on the precision of the solution, that is to say that $Re_h = \frac{(|u|+|v|)h}{\nu}$ must be "sufficiently moderate". The time step Δt is chosen so that the stability of the problem is ensured (in which the factors $r = \frac{\nu\,\Delta t}{h^2}$, $r_T = \frac{\alpha\,\Delta t}{h^2}$, $r_{C_i} = \frac{D\Delta t}{h^2}$ and $CFL = \frac{(|u|+|v|)\,\Delta t}{h} = r\,Re_h$ are involved).

For flows with large local gradients of the flow, a more appropriate definition of the "mesh Reynolds number" is that based on the velocity gradient, and evaluated using the vorticity: $Re_{h,\omega} = \frac{|\omega|\,h^2}{\nu}$. Values for quality simulations of the flow field are typically $Re_{h,\omega} \lesssim 40$ and $Re_h \lesssim 25$. You should hence monitor the maximum values of Re_h and $Re_{h,\omega}$ of your simulations, as a function of time. Here, we also have high gradients of the temperature and concentration fields, and hence we can also define "mesh Peclet numbers" based on the gradients of those fields: $Pe_{h,\nabla T} = \frac{|\nabla T|\,h^2}{\alpha}$, and $Pe_{h,\nabla C_i} = \frac{|\nabla C_i|\,h^2}{D}$. Those too should be monitored and should remain moderate for high quality simulations.

Concerning the Poisson equation, this one will be solved using the PETSc library. To install this library and get more precisions for the solving of the equation, refer to the "*PETSc installation notes*" document.

You are asked to :

1. Produce a numerical code for the study of the present configuration. The solution must be computed until a dimensionless time of $\frac{t_{end}\,U}{L_{in}} = 50$.

2. Compute and plot, as a function of $\frac{t\,U}{L_{in}}$:

   (a) The dimensionless spatially-averaged temperature, $\frac{\bar{T}-T_m}{T_m}$, with $\bar{T} - T_m = \frac{1}{L_x\,L_y}\int (T - T_m)\,dS$

   (b) The dimensionless flux of B through the outlet, $\frac{\dot{Q}_B^{up}}{C_{tot}\,U\,L_{out}}$, with $\dot{Q}_B^{up} = \int_{L_{out}} C_B\,v\,dx$

   (c) The dimensionless flux of enthalpy through the outlet, $\frac{\dot{H}}{\rho\,c\,T_m\,U\,L_{out}}$, with $\dot{H} = \int_{L_{out}} \rho_m\,c_p\,(T - T_m)\,v\,dx$

(d) The dimensionless flux of B released at the bottom surface, $\frac{\dot{Q}_B^{bottom}}{C_{tot}\,U\,L_{in}}$, with $\dot{Q}_B^{bottom} = \int_{L_x} q_B\,dx$

3. Provide also the following relevant fields: vorticity, $\frac{\omega\,L_{in}}{U}$, velocity norm, $\frac{|\mathbf{v}|}{U}$, temperature, $\frac{T}{T_m}$, and concentration of B, $\frac{C_B}{C_{tot}}$, for the dimensionless times $\frac{t\,U}{L_{in}} = 2, 4, 6, 10, 20, 30, 40, 50$.

4. Comment your results

The project can be done in a "team of two students". It can, of course, also be done alone. The report must be returned, in a recto/verso version, by **Friday 17/05/2019** at 6 pm, to one of the two teaching assistants (room A.074, Stévin building). It is also requested to upload your source code in C and your report (in .pdf format) on the Moodle website of the course. Please compress your code and your report into one single archive file before uploading it.